

```
import pandas as pd

customer_data = {
    'c_id': [1, 2, 3, 4, 5, 7, 8, 9],
    'Customer': ['Rabi', 'Raju', 'Alex', 'Rani', 'King', 'Ronn', 'Jem', 'Tom']
}

product_data = {
    'p_id': [1, 2, 3, 4, 5, 6, 7, 8],
    'product': ['Hard Disk', 'RAM', 'Monitor', 'CPU', 'Keyboard', 'Mouse', 'Motherboard', 'Power supply'],
    'price': [80, 90, 75, 55, 20, 10, 50, 20]
}

sales_data = {
    'sale_id': [1, 2, 3, 4, 5, 6, 7, 8, 9],
    'c_id': [3, 4, 3, 4, 3, 3, 2, 2, 3],
    'p_id': [3, 4, 3, 2, 3, 3, 2, 2, 3],
    'qty': [2, 1, 1, 2, 2, 3, 2, 3, 2],
    'store': ['ABC', 'DEF', 'ABC', 'DEF', 'ABC', 'DEF', 'ABC', 'DEF', 'ABC']
}

df_customer = pd.DataFrame(customer_data)
df_product = pd.DataFrame(product_data)
df_sales = pd.DataFrame(sales_data)

df_merged = df_sales.merge(df_product, on='p_id').merge(df_customer, on='c_id')
```

```
print("\n1. Products Sold:")
print(df_merged['product'].unique())
```

```
1. Products Sold:
['Monitor' 'CPU' 'RAM']
```

```
print("\n2. All Unique Products in Catalog:")
print(df_product['product'].unique())
```

```
2. All Unique Products in Catalog:
['Hard Disk' 'RAM' 'Monitor' 'CPU' 'Keyboard' 'Mouse' 'Motherboard'
 'Power supply']
```

```
print("\n3. Details of Product 'Monitor':")
print(df_product[df_product['product'] == 'Monitor'])
```

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3. Details of Product 'Monitor':
   p_id product  price
2     3  Monitor     75
```

```
print("\n4. Products with quantity > 1 in a sale:")
print(df_merged[df_merged['qty'] > 1][['product', 'qty', 'store']])
```

```
4. Products with quantity > 1 in a sale:
   product  qty store
0  Monitor     2   ABC
3     RAM     2   DEF
4  Monitor     2   ABC
5  Monitor     3   DEF
6     RAM     2   ABC
7     RAM     3   DEF
8  Monitor     2   ABC
```

```
print("\n5. Products in store ABC:")
print(df_merged[df_merged['store'] == 'ABC'][['product', 'qty']])
```

```
5. Products in store ABC:
   product  qty
0  Monitor     2
2  Monitor     1
4  Monitor     2
6     RAM     2
8  Monitor     2
```

```
print("\n6. Products in ABC with quantity > 2:")
print(df_merged[(df_merged['store'] == 'ABC') & (df_merged['qty'] > 2)])
```

```
6. Products in ABC with quantity > 2:
Empty DataFrame
Columns: [sale_id, c_id, p_id, qty, store, product, price, Customer]
Index: []
```

```
print("\n7. Details of Monitor and RAM:")
print(df_product[df_product['product'].isin(['Monitor', 'RAM'])])
```

```
7. Details of Monitor and RAM:
   p_id  product  price
1     2     RAM     90
2     3  Monitor     75
```

```
print("\n8. Total quantity sold per product:")
print(df_merged.groupby('product')['qty'].sum())
```

```
8. Total quantity sold per product:
product
CPU          1
Monitor     10
RAM          7
Name: qty, dtype: int64
```

```
df_merged['total_sale'] = df_merged['qty'] * df_merged['price']
print("\n9. Quantity and Total Sales per product:")
print(df_merged.groupby('product')[['qty', 'total_sale']].sum())
```

```
9. Quantity and Total Sales per product:
   qty  total_sale
product
CPU      1         55
Monitor  10        750
RAM       7        630
```

```
print("\n10. Quantity per product per store:")
print(df_merged.groupby(['product', 'store'])['qty'].sum().unstack(fill_value=0))
```

```
10. Quantity per product per store:
store  ABC  DEF
product
CPU      0    1
Monitor   7    3
RAM       2    5
```

```
print("\n11. Total quantity and turnover per store:")
print(df_merged.groupby('store').agg({'qty': 'sum', 'total_sale': 'sum'}))
```

```
11. Total quantity and turnover per store:
   qty  total_sale
store
ABC      9         705
DEF      9         730
```

```
print("\n12. Products not sold:")
sold_pids = df_sales['p_id'].unique()
print(df_product[~df_product['p_id'].isin(sold_pids)]['product'])
```

```
12. Products not sold:
0      Hard Disk
4      Keyboard
5      Mouse
6      Motherboard
7      Power supply
Name: product, dtype: object
```

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```
print("\n13. Customers who haven't purchased anything:")  
buying_customers = df_sales['c_id'].unique()  
print(df_customer[~df_customer['c_id'].isin(buying_customers)]['Customer'])
```

```
13. Customers who haven't purchased anything:  
0    Rabi  
4    King  
5    Ronn  
6     Jem  
7     Tom  
Name: Customer, dtype: object
```